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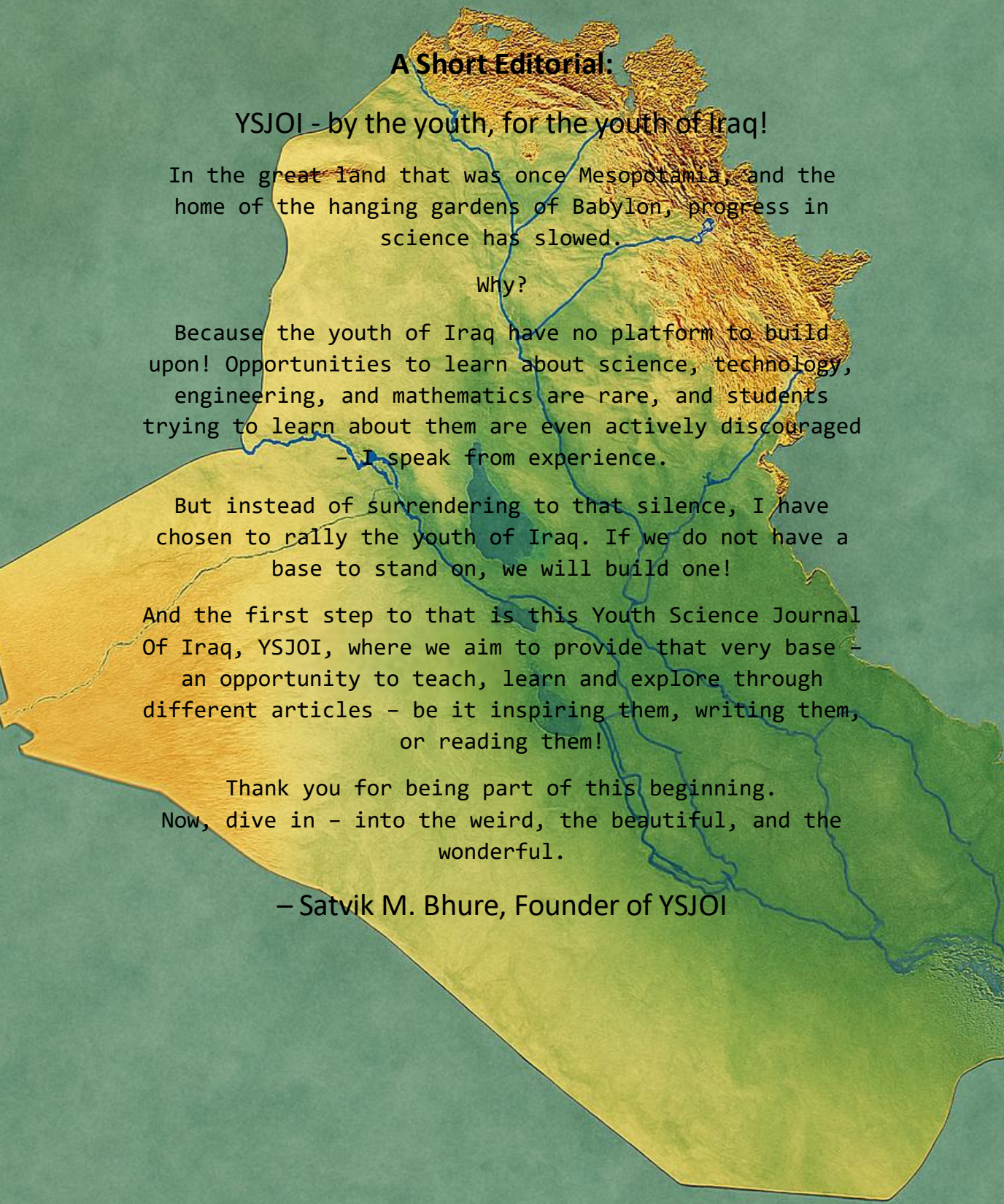
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A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: How are we sure that the Earth is round?

– Saif Mariwan, 13

The Earth's spherical shape is a scientific truth. While the ground looks flat to a human observer, this is a mere illusion of scale. We can easily prove the planet is a sphere using simple geometry and everyday observations without needing satellite imagery.

A clear proof of curvature happens at the coastline. When a ship sails away, it does not just shrink. The hull vanishes beneath the water first, and the mast disappears last. If you quickly climb a hill or a tower, the ship comes back into view. This happens because higher altitude extends your line of sight over the physical curve of the ocean.

Looking at the sky provides global proof. During a lunar eclipse, the Earth blocks the Sun and casts a shadow on the Moon. For millennia, observers have noted that this shadow is always a perfect arc. A sphere is the only shape that casts a circular shadow from every single angle. If the Earth were a flat disc, its shadow would distort into a thin line or oval as the Sun shifted underneath it.

Traveling north or south also changes the night sky. If the Earth were flat, everyone would see the identical stars. Instead, the North Star sinks lower as you travel toward the equator and vanishes entirely in the Southern Hemisphere. Meanwhile, entirely new constellations, like the Southern Cross, appear. This shifting view proves you are walking along a continuous curve.

In 240 BCE, the scholar Eratosthenes used simple geometry to calculate the Earth's size. At noon on the summer solstice, the Sun shone directly down a well in Syene, casting no shadow. In Alexandria, a stick cast a shadow at an angle of 7.2 degrees, which is one-fiftieth of a circle. He multiplied the distance between the cities by fifty and found the global circumference with incredible accuracy.

Time zones exist because the sun cannot light up both sides of a ball at once. Sunsets also prove this; the sun sinks below the horizon from the bottom up while keeping its exact same size. Furthermore, large weather systems like hurricanes rotate counterclockwise in the north and clockwise in the south due to the Coriolis effect. Because a sphere is wider at the equator than the poles, different latitudes spin at different speeds, creating this multi-directional rotation.

Question II: Why are constellations important?

– *Faisal Safwan, 14*

In many movies, captains of small ships close one eye, put a hand up to the sky, and align it with some stars; then they somehow know where they are and where to go. But it's not all fiction.

Let's assume that a captain in the Northern Hemisphere wants to find true north. The first thing he has to do is look for Ursa Major (The Big Bear), seeking the seven stars that make the bowl-shaped part specifically.

Then, the captain must find the two stars on its outer edge, opposing the 'handle.' Those two stars are called Merak (the bottom one) and Dubhe (the top one). This is where the hand placement comes in.

The captain aligns his hand with the line between Merak and Dubhe (let's call it line A), getting an estimate of its length. The captain multiplies this length by five and continues line A, adding this length. This should give the captain a rough estimate of the position of Polaris, the North Star. This allows him, after another series of similar techniques, to find True North.

Now, beyond wayfinding, what other purposes do constellations have?

Firstly, they were used to track time at night. During the day, sundials were used. At night, they used the positions of constellations and compared them to their original positions at sunset.

Another common use was to form agricultural calendars, as a way of organizing when to reap and when to sow based on which constellation shows up over the horizon. Of course, this changes every month or so because the Earth doesn't spin perfectly on its axis. In fact, it somewhat wobbles, so the constellations that show up are slightly shifted in position each time.

Of course, the use of constellations in calendars is not entirely abolished, considering we still recognize Zodiac signs. They were developed based on a pattern noticed by ancient astronomers over 2000 years ago, when they saw that the sun passes through a certain constellation each 30-day month. They named these constellations and set their specific dates. However, due to the wobbling mentioned earlier, the sun no longer seems to pass through the same constellations on the same dates.

Of course, this is a mere simplification of the importance of constellations. They shaped our past, and our understanding of geography and space irreparably.

Question III: What are the oldest signs of life on Earth?

– Christina Andrews, 17

Life on Earth has developed an extraordinary amount from how it first started here on our planet Earth. So different, indeed, that today people spend a lot of time think about how life first was, and searching for the oldest signs of its earliest existence. But how could a researcher find such information? And how could they prove it?

As of right now there are several claims of what the first life form was, each with a different type of evidence to support it. Only one, however, was widely accepted. The earliest recorded biological fossil is called a stromatolite, found in Australia in old rocks named geyserite and is estimated to be around 3.480 billion years old. Many other small fossilized microorganisms have also been found around the 3.4 billion year old range, including rocks from the oceans of South Africa all the way back to Australian lands. Going forward roughly 1.7 billion years forward in time shows us preserved compounds which could be biologic life forms that relied on aerobic respiration unlike all the previously mentioned organisms.

Lastly, there is potentially a finding that could push the ceiling of the oldest life form further back to as old as 4.1 billion years ago. Separated graphite in a metallic ore has been found and the graphite is believed to have come from an organism that lived around that time.

Unfortunately, the graphite was only found in a singular grain of this ore which is why this is not a confirmed oldest sign of life on Earth.

With this information now we see that life on Earth could have appeared along with the formation of the oceans, since the oceans started forming around 4.4 billion years ago and they only started covering a large surface and creating stable expanses of water around 3.8 billion years ago, and so this new organism may have lived when the average temperature of Earth was still above the boiling point of water.

Question IV: Where is gravity weakest and how can we use this weakness?

– Omed Fikri, 12

Gravity is one of the forces we experience every day; however, its strength is not the same everywhere on Earth. It is slightly weaker at the equator and stronger at the poles. Although this difference is small, it can have some interesting practical uses, especially when it comes to space travel and studying our planet.

As the Earth spins, it creates an outward effect that slightly reduces the force of gravity around the equator, and because the Earth is wider at the equator, people there are farther away from the center. Therefore, gravity is weakest at the equator mainly because of our planet's shape and rotation.

This weakness is particularly useful when launching rockets into space. Launch sites near the equator can take advantage of the Earth's rotation as it gives them an extra boost as they begin their journey. Because the rocket already has some speed from the spinning Earth, less fuel and energy may be needed to reach space, which makes launches more efficient and can allow more useful equipment to be carried along.

Differences in gravity are also useful to scientists. By carefully measuring gravity in different locations, scientists can learn more about what lies beneath the Earth's surface, such as underground rocks, minerals, and structures that cannot easily be seen or identified from above.

The weakness of gravity at the equator may seem like a small detail, but it has real value. It can support more efficient rocket launches and provide clues about what's below the Earth's crust. Who knew such a small advantage could push the limits of space exploration and planet research!

Beyond space travel and geology, variation in gravity means we can graph the oceans and monitor our climate. By watching with satellites the way that gravity shifts with time and geography, we can monitor large scale environmental changes.

Question V: How fast is the universe expanding?

– *Ahmed Alsawaf, 1*5

The universe has been expanding continuously since the Big Bang. While its total size remains unknown and could potentially be infinite, astronomers confirm this expansion through redshift.

Redshift occurs when the wavelengths of light emitted from distant celestial bodies stretch into longer, redder wavelengths by the time they reach our instruments, directly indicating that space itself is stretching out.

Crucially, the universe lacks a physical center. It is expanding uniformly in all directions, but there is no single, fixed speed for this expansion.

Instead, the velocity at which an object appears to recede depends directly on its distance from the observer.

To determine this recession speed at a specific distance, cosmologists use a foundational value known as the Hubble constant.

The Hubble constant quantifies the precise relationship between a galaxy's recession speed and its distance from Earth. To apply this constant, distances are typically measured in megaparsecs (Mpc), where a single megaparsec equals approximately 3.26 million light-years. The currently accepted value for the Hubble constant is roughly 70 kilometers per second per megaparsec.

This measurement means that for every megaparsec of space separating an observer from an object, the apparent speed of that object moving away increases by about 70 kilometers per second.

For example, if a galaxy is situated 10 Mpc away from Earth, its recession speed due to cosmic expansion is approximately 700 km/s.

If another galaxy lies twice as far, at 20 Mpc, it moves away twice as fast, at roughly 1,400 km/s.

Because expansion scales directly with distance, there is no universal "speed limit" for the universe's growth as a whole.

Rather, space expands everywhere at once, causing galaxies that are farther apart to pull away from one another at proportionately higher speeds.

Acknowledgements

I (Satvik M. Bhure) have many people to thank for this incredible piece of work!

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And we thank you, reader, for supporting our mission to help the Iraqi youth where there has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at https://YSJOI.is-great.org

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